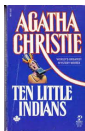
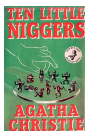


AI, computational social choice, and democracy: ten little talks (a tribute to Agatha Christie)

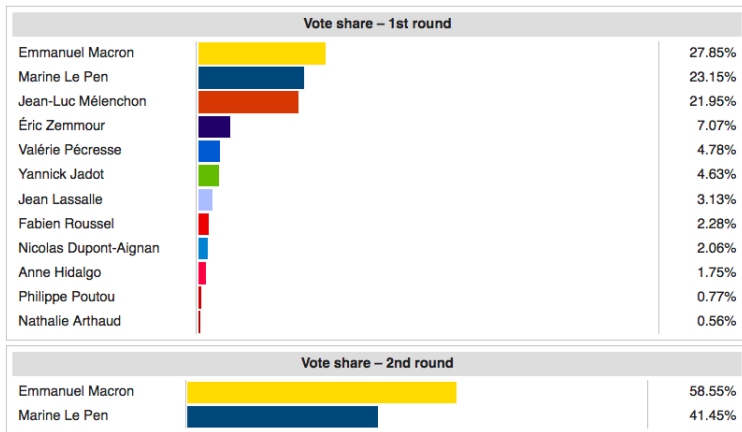
Jérôme Lang
CNRS & Université Paris-Dauphine PSL
lang@lamsade.dauphine.fr

September 25, 2026



Social choice theory

Designing and analysing methods for collective decision making



Social choice theory

Designing and analysing methods for collective decision making

 [HOME](#) [NEW POLL](#) [DEVELOPERS](#) [CONTACT US](#) [jeromelang](#) 

A restaurant for tonight

5 voters have participated in this poll.

[Vote](#) [Results](#)

		japanese	indian	tunisian	pizza	crêperie
 	Bob	0	0	-	+	+
 	Carol	+	+	++	-	-
 	David	-	0	0	+	0
 	Edith	+	+	++	-	0
 	Ann	++	++	+	--	-

The poll is opened until **March 25, 2025** (unless the poll creator decides to close it before).

[Vote](#) [Results](#) [Settings](#) [Data page](#)

Social choice theory

Designing and analysing methods for collective decision making



A very rough history of social choice

1. 18th century: Condorcet and Borda (first Economics and AI day, 14 July 1789)
2. 1951: birth of social choice theory (economics/mathematics); mostly **axiomatic** results such as **impossibility theorems** (most celebrated: Arrow's)
3. from the 1990's: **computational turn**.

(Computational) social choice



AI and Computational Social Choice

AI / CS have contributed to **reshape** social choice:

- ▶ new techniques
- ▶ new paradigms
- ▶ new objects of study, new applications
- ▶ bottom-up approaches

Social Choice Rules

- **input**: agents express preferences over alternatives/candidates
- **output**: an alternative

Choose the temperature in the room? **Various input formats**

Ann: 17

Bob: 20

Carol: 19

David: 17

uninominal

Ann: $17 \succ 18 \succ 19 \succ 20$

Bob: $20 \succ 19 \succ 18 \succ 17$

Carol: $19 \succ 20 \succ 18 \succ 17$

David: $17 \succ 18 \succ 19 \succ 20$

ordinal

	17	18	19	20
Ann	+	+	+	
Bob				+
Carol		+	+	+
David	+	+		

approvals

	17	18	19	20
Ann	50	30	20	0
Bob	0	0	0	100
Carol	0	40	50	10
David	40	30	20	10

evaluations

Choosing topics

<https://whale5.imag.fr/polls/f58cd0e1-6462-4c66-98a4-90eb34631184>



1. Liquid democracy

Forms of democracy

- ▶ **Representative democracy**: citizens elect representatives to make decisions on their behalf.
- ▶ **Direct democracy**: citizens govern directly and express their opinion on any issue.
- ▶ **Deliberative democracy**: decisions emerge from inclusive deliberation, e.g., citizens' assemblies.
- ▶ **Participatory democracy**: citizens are involved not just at election time but continuously, e.g., participatory budgeting.

1. Liquid democracy

- ▶ **Representative democracy**: citizens elect representatives to make decisions on their behalf.
- ▶ **Liquid/fluid democracy**: citizens can choose either to vote on an issue, or to delegate to someone else.
- ▶ **Direct democracy**: citizens govern directly and express their opinion on any issue.

1. Liquid democracy

Committee election

Who should be elected at the
new steering board?

Do you want to vote yourself or del-
egate your vote to a trusted peer?

Classical social choice
Aggregating *preferences*
No ground truth

1. Liquid democracy

English idioms

You will be given English idioms, and asked to identify their meaning.



Do you want to vote yourself or delegate your vote to a trusted peer?

Landmarks

You will be shown pictures of landmarks, and asked to say in which country they are.



don't delegate

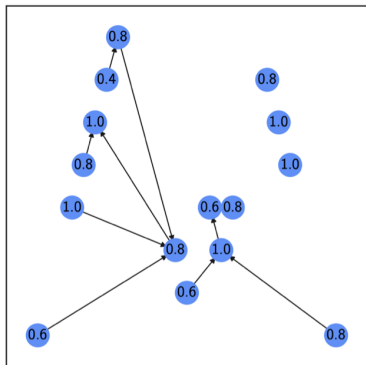
Do you want to vote yourself or delegate your vote to a trusted peer?

Epistemic social choice: Aggregating *beliefs* about a *ground truth*

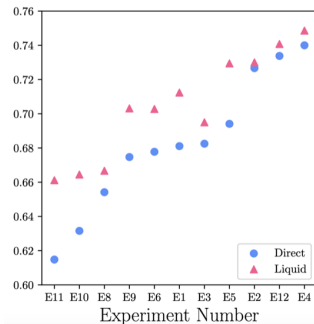
1. Liquid democracy

English idioms

You will be given English idioms, and asked to identify their meaning. Do you want to vote yourself or delegate your vote to a trusted peer?



Delegation graph



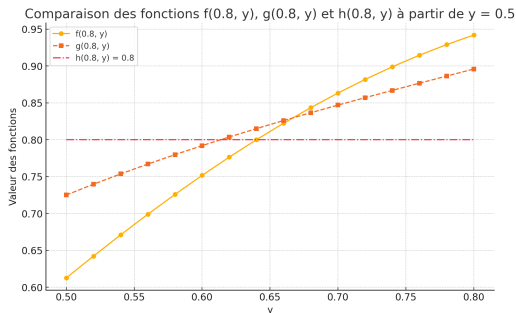
Accuracy

Source: Manon Revel

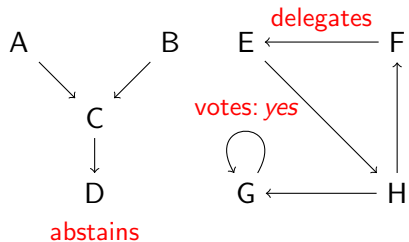
1. Liquid democracy: delegation can harm

- ▶ one expert E, giving a correct answer with probability $x = 0.8$
- ▶ four non-experts, giving a correct answer with independent probabilities $y \in [0.5, x]$ (or delegating to E)

	probability that majority is correct
no delegation	$f(x, y) = 6xy^2(1 - y)^2 + 4y^3(1 - y) + y^4$
one delegation $\rightarrow$ E	$g(x, y) = x(1 - (1 - y)^3) + (1 - x)y^3$
$\geq$ two delegations $\rightarrow$ E	$h(x, y) = x$

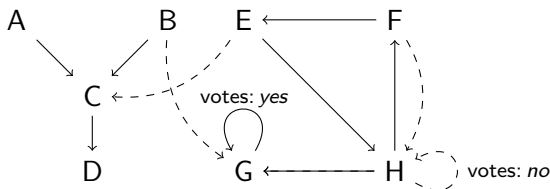


1. Liquid democracy: cycles and sinks



Cycles?
Delegations leading nowhere?

→ Ranked delegations

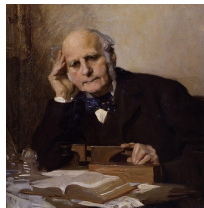


2. Epistemic Voting

- ▶ there is a hidden ground truth
- ▶ votes are noisy signals about this ground truth
- ▶ aggregating votes sometimes allows to uncover the truth with high probability

Wisdom of the crowds: Galton's ox

- ▶ 1906 country fair in Plymouth: 800 people participated in a contest to estimate the weight of an ox.
- ▶ the median guess, 1207 pounds, was accurate within 1% of the true weight of 1198 pounds



2. Epistemic Voting

Condorcet's jury theorem:

- ▶ two possible outcomes (e.g., yes/no), one of them is correct (ground truth)
- ▶ n voters; each voter has a probability $p > \frac{1}{2}$ of voting for the correct decision.
- ▶ votes are independent

Then the probability that the majority decision is correct increases with n and tends to 1 when n tends to infinity

Extensions: more than two outcomes, voters with different expertise, correlated votes...



2. Epistemic Voting and Crowdsourcing: Approval Voting



Crowdsourcing via approval voting

In which of the 20 districts of Paris was this picture taken? You may give several answers. You will get a reward if your selection contains the true answer, minus a penalty that increases with the size of your selection.

2. Epistemic Voting and Crowdsourcing: Approval Voting

	12	13	14	15	16	17	18	19	20	expertise?
Ann							+			
Bob			+		+			+	+	
Carol		+		+		+		+		
David							+		+	
Eva			+	+	+	+	+	+	+	
Fred	+									
Gloria					+		+	+	+	
#	2	2	2	2	3	2	4	4	4	

2. Epistemic Voting and Crowdsourcing: Approval Voting

Crowdsourcing via approval voting

	12	13	14	15	16	17	18	19	20	expertise?
Ann							+			high
Bob			+		+			+	+	med-
Carol		+		+		+		+		med-
David							+		+	med+
Eva			+	+	+	+	+	+	+	low
Fred	+									high?
Gloria					+		+	+	+	med-
#	2	2	2	2	3	2	4	4	4	

2. Epistemic Voting and Crowdsourcing: Approval Voting

Crowdsourcing via approval voting

	12	13	14	15	16	17	18	19	20	expertise?
Ann							+			high
Bob			+		+			+	+	med-
Carol		+		+		+		+		med-
David							+		+	med+
Eva			+	+	+	+	+	+	+	low
Fred	+									low!
Gloria					+		+	+	+	med-
#	2	2	2	2	3	2	4	4	4	

2. Epistemic Voting and Crowdsourcing: Approval Voting

	12	13	14	15	16	17	18	19	20	expertise?
Ann							+			high
Bob			+		+			+	+	med-
Carol		+		+		+		+		med-
David							+		+	med+
Eva			+	+	+	+	+	+	+	low
Fred	+									low!
Gloria					+		+	+	+	med-
#							•			

Statistical machine learning approach:

- ▶ alternatives approved by many competent individuals are more likely
- ▶ individuals approving few alternatives, some of them likely, are more competent
- ▶ works well on experimental datasets

2. Epistemic Voting: Surprisingly Popular Answers

Question 1 Which of Rome and New York City is farther North?

2. Epistemic Voting: Surprisingly Popular Answers

Question 1 Which of Rome and New York City is farther North?

- ▶ usually 70 % of respondents say New York
- ▶ latitudes: Rome: 41.87 N, New York 40.43 N
- ▶ majority is wrong, wisdom of the crowds fail...

2. Epistemic Voting: Surprisingly Popular Answers

Question 1 Which of Rome and New York City is farther North?

- ▶ usually 70 % of respondents say New York
- ▶ latitudes: Rome: 41.87 N, New York 40.43 N
- ▶ majority is wrong, wisdom of the crowds fail...

Question 2 Guess how many people give the same answer as you

answer	number of votes	guess
Rome	30	25
New York	70	90

Taking account of Question 2 allows to uncover the ground truth!

D. Prelec, A Bayesian Truth Serum for Subjective Data, *Science*, 2004

3. Deliberative democracy : representative mini-publics



Citizens assemblies in the world:
Data OCDE 2020
Sortition foundation:
<https://www.sortitionfoundation.org/where>



Réunion publique, Saint-Etienne Centre,
Septembre 2023. Photo Municipalité



Concertation Chambéry Métropole Coeur des
Bauges, Mai 2017. Photo Métropole

Thanks: Antoinette Baujard

3. Deliberative democracy : representative mini-publics

Selecting mini-publics... how?

- ▶ Some citizens are randomly selected
- ▶ Some of them accept to participate, others decline
- ▶ And we get a ... **non-representative** selection of citizens

3. Deliberative democracy : representative mini-publics

The aim is to create a group of 20 participants who are representative on a number of socio-demographic attributes:

- ▶ women 50% (9-11), men 50% (9-11)
- ▶ age 18-25: 15% (3), 25-45: 30% (5-7), 45-65: 30% (5-7), 65+: 25% (4-6)
- ▶ city centre 20% (3-5), North 20% (3-5), South 20% (3-5), East 20% (3-5), West 20% (3-5)
- ▶ students 10% (2), executives: 20% (3-5), employees: 20% (3-5), workers: 20% (3-5), unemployed: 10% (2), retired: 20% (3-5)
- ▶ etc.

3. Selecting representative mini-publics: what should **not** be done

1. Call for volunteers and select the first 20.

W, age 22, centre, student	M, age 26 North, executive
W, age 30, East, executive	W, age 68, centre, retired
M, age 56, centre, executive	M, age 20, East, student
...	...

2. Select people randomly and ask them if they agree, until we have 20 people.

W, age 22, centre, student	M, age 45, South, worker
M, age 67, East, retired	W, age 38, centre, employee
W, age 50, West, unemployed	W, age 34, centre, executive
W, age 74, North, retired	M, age 19, centre, student
M, age 82, centre, retired	W, age 54, West, employee
M, age 29, South, unemployed	M, age 55, North, executive
...	...

3. Selecting representative mini-publics: what can sometimes be done

3. Stratification

- ▶ with a single attribute (W/M) : build two mini-publics of 10 persons each with one of the previous algorithms ✓
- ▶ with two attributes : ✓

W 18-25: 7.5% (1-2)	H 18-25 : 7.5% (1-2)	W 25-45 : 15% (3)
M 25-45 : 15% (3)	F 45-65 : 15% (3)	M 45-65 : 15% (3)
W 65+ : 15% (3)	M 65+ : 10% (2)	

- ▶ with 4 attributes : **infeasible!**

W 18-25 C student: 5%	W 18-25 C worker: 0.5%
W 18-25 C employee: 1%	W 18-25 C unemployed: 1%
W 18-25 S student: 3%	W 18-25 S worker: 1%
W 18-25 S employee: 2%	W 18-25 S unemployed: 1.5%
W 18-25 N student: 3%	W 18-25 N worker: 0.5%
W 18-25 N employee: 3%	W 18-25 N unemployed: 1.5 %
...	...
...	...

3. Selecting representative mini-publics: doing better

4. Randomization followed by dynamic selection

randomly selected person	ask them?	do they accept?
W, 22, centre, student	yes	yes
M, 45 , South, worker	yes	no
W, 38 , centre, executive	yes	yes
W, 50 , West, unemployed	yes	no
W, 74 , North, retired	yes	yes
M, 19 , centre, student	no	—
M, 82 , centre, retired	yes	no
W, 54 , North, employee	yes	yes
M, 29 , South, unemployed	yes	no
W, 55 , North, unemployed	no	—
M, 33 , centre, executive	no	—
etc.		

needs lots of time and interactions; possible for mini-minipublics,
not for large citizen assemblies

3. Selecting representative mini-publics: doing even better

5. Fair and representative lottery

- ▶ everyone is asked if they volunteer or not.
- ▶ we compute a lottery in which each volunteer has a probability of being selected, so as:
 - ▶ to correct acceptance biases;
 - ▶ to ensure that every volunteer has the fairest possible chance of being selected.
- ▶ simple example with a single attribute

age	% population	# volunteers	need	probability
18-25	15%	20	3	0.15
25-45	30%	18	6	0.33
45-64	30%	12	6	0.5
25-45	25%	6	5	0.80

3. Selecting representative mini-publics: doing it really right

5. Fair sortition

Step 1 The algorithm constructs an **optimal lottery** over mini-publics guaranteeing that

- ▶ representativity constraints are satisfied
- ▶ each volunteer has a reasonable probability to be selected [maximize the minimum probability of being selected, over all the volunteers]
- ▶ this lottery is made public

Step 2 From this optimal lottery, a minipublic is **drawn randomly**.

3. Selecting representative mini-publics: doing it really right

Determining the optimal lottery is very difficult - impossible without an efficient program.

<https://panelot.org/>



Panelot

1. Compute a Civic Lottery

2. Draw a Panel

[Learn More](#)

Everyone deserves a fair chance

Panelot (panel selection by lot) is a not-for-profit system for randomly selecting citizens' panels in a way that is representative of the population and fair to volunteers.



Step 1: Compute a Civic Lottery

Generate a fair distribution over panels, each of



Step 2: Draw a Panel

Select a single panel from a precomputed civic



The Sortition Foundation

<https://www.sortitionfoundation.org/>

4. Social Choice and Moral AI

- ▶ Example 1: Autonomous vehicles

<https://www.moralmachine.net/>

trolley problem (→ autonomous cars):

Awad, Dsouza, Bonnefon, Shariff, Rahwan: Crowdsourcing moral machines. Commun. ACM 63(3): 48-55 (2020)

- ▶ Example 2: Allocating scarce medical resources

- ▶ How to allocate artificial ventilators to Covid patients when demand exceeds supply?
- ▶ Kidneys transplant: how to choose among potential receivers?

Freedman, Schaich Borg, Sinnott-Armstrong, Dickerson, Conitzer: Adapting a kidney exchange algorithm to align with human values. Artif. Intell. (2020)

4. Moral AI: Learning Societal Preferences

Allocating ventilators to Covid patients when demand exceeds supply

patient	rank in queue	age	expected # years gained	gained probability of survival	
Ann	1	80	10	0.9	
Bob	2	60	20	0.9	
Carol	3	50	25	0.8	life imprisoned for murder
David	4	70	15	0.7	past kidney donator
Elena	5	65	15	0.6	doctor
Freddie	6	55	23	0.7	a doctor's partner
Gautam	7	45	16	0.4	

4. Moral AI: Learning Societal Preferences

Allocating ventilators when demand exceeds supply: four principles:

1. **utilitarianism**: maximise the sum of individual satisfactions (social welfare).
 - 1.1 maximise the expected number of lives saved
 - 1.2 maximiser the expected number of years of lifes saved
2. **ex ante egalitarianism**: same chances *a priori* to access resources
 - 2.1 first come, first serve
 - 2.2 random allocation with uniform probability
3. **reward and merit**: for instance, give priority to health care workers. Two interpretations:
 - ▶ reward for past or future actions
 - ▶ maximisation of instrumental value
4. **ex post egalitarianism**, or compensation: priority to patients who have been most disadvantaged until now (i.e., who would have the shortest lives if they are not allocated a ventilator).

4. Moral AI: Learning Societal Preferences

Question. How can we infer normative trade-offs from past decisions?

Setup. For each decision i , we observe two candidates x^i, y^i with features $(x_1^i, \dots, x_q^i), (y_1^i, \dots, y_q^i)$

Model. Interpretable linear social score

$$S(x) = \sum_j w_j x_j$$

Preference: choose x^i over y^i iff $S(x^i) > S(y^i)$.

Learning (from past choices). Constrained logistic regression

Output. w summarizing societal priorities

4. Social Choice and Moral AI

Learning societal preferences

- ▶ whom individuals should we query?
 - ▶ doctors?
 - ▶ politicians?
 - ▶ the citizens?
- ▶ who chooses the aggregation method?
- ▶ which applications?
 - ▶ **kidney exchange**: Freedman, Schaich Borg, Sinnott-Armstrong, Dickerson, Conitzer: Adapting a kidney exchange algorithm to align with human values. *Artif. Intell.* (2020)
 - ▶ **trolley problem (→ autonomous cars)**: Awad, Dsouza, Bonnefon, Shariff, Rahwan: Crowdsourcing moral machines. *Commun. ACM* 63(3): 48-55 (2020)
 - ▶ **matching student to university programs / schoolchildren to schools**
 - ▶ etc.

5. Social choice engineering at Université Paris-Dauphine



- ▶ huge construction works in the whole building 2022-2027
- ▶ one building, 600 offices, most occupied by one or two persons
- ▶ $> 90\%$ of the building will be completely rebuilt
- ▶ 5 big phases, whose duration is known with some uncertainty
- ▶ it is known which offices will be unavailable at each phase
- ▶ initial office allocation known, final state (almost) known
- ▶ people moving in average twice + possible compression at some intermediate phase

Students: this should not prevent you from coming and studying with us!

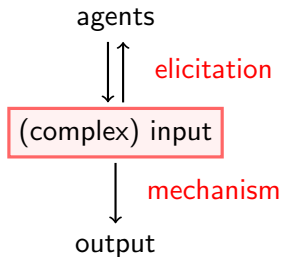
5. Social choice engineering at Université Paris-Dauphine

- ▶ the university asked us to help finding a fair and efficient reallocation sequence
- ▶ expertise needed in AI, OR and social choice
- ▶ a fair division problem? Yes but:
 - ▶ 6 research labs + teaching departments + central services
⇒ not clear who the agents are: individuals, groups, both?
 - ▶ heavily non-additive preferences: desire for labs/departments to remain grouped, for moves to be timewise not too close, ...
 - ▶ uncertainty

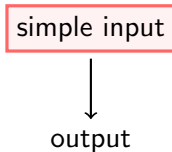
temporal fair division problem with individual and group fairness, complex nonadditive preferences and uncertainty!

- ▶ each of these complications has been studied individually
- ▶ no known framework / algorithm for our problem
- ▶ **social choice engineering!** (here and elsewhere)

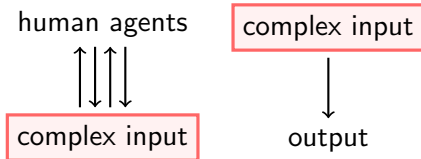
5. Social Choice Engineering



What we know how to do



What is missing



6. Surprise topic

6. Surprise topic

... Sorry, unavailable

6. Surprise topic

... Sorry, unavailable

Possible causes:

- ▶ I did not have time to write the slides
- ▶ a law just passed that prohibits me to talk about this topic
- ▶ someone talked about it already

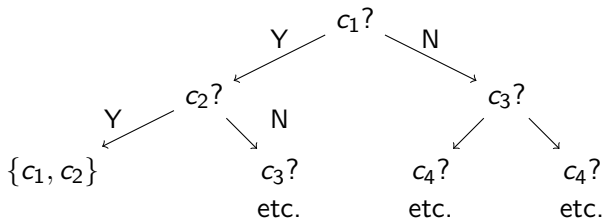
More generally: candidates sometimes are also unavailable.

6. Voting with (possibly) unavailable candidates

- ▶ a university department wants to hire two new professors
- ▶ three research groups in the department
- ▶ group 1 (size $3/8$) likes c_1, c_3
- ▶ group 2 (size $3/8$) likes c_1, c_4
- ▶ group 3 (size $1/4$) likes c_2
- ▶ **Proportional Approval Voting:**

$$\{c_1, c_2\} \succ \{c_3, c_4\} \succ \{c_2, c_3\} \sim \{c_2, c_4\}$$

- ▶ **Policy:**



7. Complex alternatives → Combinatorial domains

- ▶ there are several possible topics I can speak during my talk
 - ▶ I have time to talk only about two topics
 - ▶ Ann: would like to hear about t_1 or t_3 , and about t_2 or t_4 .
 - ▶ Bob: likes t_1 and t_4 , and in case t_1 is not selected then t_2 .
 - ▶ Carol: likes t_3 and that's all.
-
- ▶ focus on preferential dependencies (and how to let voters express them)

7. Complex alternatives → Multiwinner elections

We can now select three topics. The votes of the attendees:

	t_1	t_2	t_3	t_4	t_5
8 voters	+	+	+		
3 voters				+	
1 voter					+

Three possible criteria → three families of rules

excellence	t_1, t_2, t_3
diversity	t_1, t_3, t_4
proportionality	t_1, t_2, t_5

7. Complex alternatives → Multiwinner elections

We can now select three topics. The votes of the attendees:

	t_1	t_2	t_3	t_4	t_5
8 voters	+	+	+		
3 voters				+	
1 voter					+

Three possible criteria → three families of rules

excellence	t_1, t_2, t_3
diversity	t_1, t_3, t_4
proportionality	t_1, t_2, t_5

7. Complex alternatives → Multiwinner elections

We can now select three topics. The votes of the attendees:

	t_1	t_2	t_3	t_4	t_5
8 voters	+	+	+		
3 voters				+	
1 voter					+

Three possible criteria → three families of rules

excellence	t_1, t_2, t_3
diversity	t_1, t_3, t_4
proportionality	t_1, t_2, t_5

7. Complex alternatives → Multiwinner elections

We can now select three topics. The votes of the attendees:

	t_1	t_2	t_3	t_4	t_5
8 voters	+	+	+		
3 voters				+	
1 voter					+

Three possible criteria → three families of rules

excellence t_1, t_2, t_3

diversity t_1, t_3, t_4

proportionality t_1, t_2, t_5

- focus on properties, especially proportionality

7. Complex alternatives → Multiwinner elections

Multiwinner voting rules

The form of the input can be one of the following

- ▶ **party-list elections**: express preferences over parties or attributes
 - ▶ apply an apportionment method
- ▶ **multi-winner elections**: express preferences directly over candidates. Preferences expressed by voters can be:
 - ▶ naming a single candidate
 - ▶ **naming a subset of candidates**
 - ▶ **ranking the candidates**
 - ▶ assigning scores to candidates

7. Complex alternatives → Multiwinner elections

Multiwinner voting rules with approval input

	t_1	t_2	t_3	t_4	t_5
8 voters	+	+	+		
3 voters				+	
1 voter					+

Thiele rules

- ▶ $w = (w_1, \dots, w_k)$ — here $k = 3$
- ▶ if $|S \cap A_i| = r$ then A_i gives a score $w_1 + \dots + w_r$ to S
- ▶ **standard approval voting**: $w = (1, 1, 1, 1) \mapsto \{t_1, t_2, t_3\}$
- ▶ **approval-based Chamberlin-Courant**:

$$w = (1, 0, 0, 0) \mapsto \{t_1, t_4, t_5\}$$

- ▶ **proportional approval voting (PAV)** (Thiele, 1895):

$$w = (1, 1/2, 1/3, 1/4) \mapsto \{t_1, t_2, t_4\}$$

$$t_1 t_2 t_4 : 8 \frac{3}{2} + 3 = 15; \quad t_1 t_2 t_3 : 8 \frac{11}{6} = \frac{44}{3}; \quad t_1 t_4 t_5 : 8 + 3 + 1 = 12$$

Non-Thiele rules: Phragmen, Equal Shares ... 

7. Complex alternatives → Participatory budgeting

- ▶ topics now have durations
- ▶ total budget: 30 minutes

	t_1	t_2	t_3	t_4	t_5	t_6
100×	+	+				
90×			+			
30×				+	+	+
30×				+	+	
10×	+			+		
<i>cost</i>	9	9	9	4	4	4

7. Complex alternatives → Participatory budgeting

- ▶ topics now have durations
- ▶ total budget: 30 minutes

	t_1	t_2	t_3	t_4	t_5	t_6
$100\times$	+	+				
$90\times$			+			
$30\times$				+	+	+
$30\times$				+	+	
$10\times$	+			+		
<i>cost</i>	9	9	9	4	4	4

A more common interpretation:

- ▶ $t_1, \dots, t_6$ are projects with costs
- ▶ total budget: 30 M€

7. Complex alternatives → Participatory budgeting

	t_1	t_2	t_3	t_4	t_5	t_6
100×	+	+				
90×			+			
30×				+	+	+
30×				+	+	
10×	+			+		
cost	9	9	9	4	4	4

available budget: 30

The greedy method

topic	#votes	cost	
t_1	110	9	●
t_2	100	9	●
t_3	90	9	●
t_4	70	4	
t_5	60	4	
t_6	30	4	

Good?

7. Complex alternatives → Participatory budgeting

	t_1	t_2	t_3	t_4	t_5	t_6
100×	+	+				
90×			+			
30×				+	+	+
30×				+	+	
10×	+			+		
<i>cost</i>	9	9	9	4	4	4

available budget: 30

<i>topic</i>	<i>#votes</i>	<i>cost</i>		
t_1	110	9	●	●
t_2	100	9	●	
t_3	90	9	●	●
t_4	70	4		●
t_5	60	4		●
t_6	30			●

Need to ensure fairness to groups of voters through proportionality

7. Complex alternatives → Judgment aggregation

We can select three topics. The votes of the attendees:

	t_1	t_2	t_3	t_4	t_5
5 voters	+	+	+		
3 voters	+	+			+
1 voter				+	+
1 voter			+		+
2 voters				+	

Admissible committees are those that satisfy the constraint

$$(t_1 \vee t_3) \wedge (t_2 \vee t_5) \wedge \neg(t_1 \wedge t_4 \wedge t_5) \wedge \neg(t_2 \wedge t_4 \wedge t_5) \wedge (t_3 \rightarrow t_4)$$

- focus on complex feasibility constraints

7. Complex alternatives

focus on	proportionality guarantees	complex preferences	complex constraints
combinatorial domains		+	
multiwinner elections	+		
participatory budgeting	+		(+)
judgment aggregation			+

8. Fair Division of Indivisible Items

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>Ann</i>	15	3	2	2	6
<i>Bob</i>	7	5	5	5	7
<i>Carol</i>	20	3	3	3	3

- ▶ $v_{Bob}(b) = 5 =$ value of item b for Bob
- ▶ Assume agents have *additive* valuations:

$$v_{Bob}(\{b, e\}) = 5 + 7 = 12$$

- ▶ *envy-freeness* (EF): every agent i weakly prefers her share to the share of any other agent j
- ▶ Ann prefers Bob's share $\{b, e\}$ to her own $\{c, d\}$: the blue allocation is not envy-free
- ▶ There is no envy-free allocation!

8. Fair Division of Indivisible Items

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>Ann</i>	15	3	2	2	6
<i>Bob</i>	7	5	5	5	7
<i>Carol</i>	20	3	3	3	3

- ▶ A weakening of EF: *envy-freeness up to one good* (EF1):
- ▶ The blue allocation is EF1:
 - ▶ Ann no longer envies Bob if we remove one good from Bob's share: $v_{Ann}(\{b, e\} \setminus \{e\}) = 3 \leq v_{Ann}(\{c, d\}) = 4$
 - ▶ Ann no longer envies Carol if we remove one good from Carol's share: $v_{Ann}(\{a\} \setminus \{a\}) = 0 \leq v_{Ann}(\{c, d\}) = 4$
 - ▶ Bob and Carol do not envy anyone.
- ▶ An EF1 allocation is guaranteed to exist (for additive valuations) and can be computed in polynomial time.

8. Fair Division of Indivisible Items

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>Ann</i>	15	3	2	2	6
<i>Bob</i>	7	5	5	5	7
<i>Carol</i>	20	3	3	3	3

- ▶ Between EF1 and EF: *envy-freeness up to any good* (EFX)
- ▶ Ann still envies Bob if we remove *b* from Bob's share:
 $v_{Ann}(\{b, e\} \setminus \{b\}) = 6 > v_{Ann}(\{c, d\}) = 4$
- ▶ the blue allocation is not EFX.

8. Fair Division of Indivisible Items

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>Ann</i>	15	3	2	2	6
<i>Bob</i>	7	5	5	5	7
<i>Carol</i>	20	3	3	3	3

- ▶ Between EF1 and EF: *envy-freeness up to any good* (EFX)
- ▶ the red allocation is EFX: Bob envies Ann but removing any good from Ann's share eliminates this envy; and similarly for Ann's envy towards Carol.

8. Fair Division of Indivisible Items

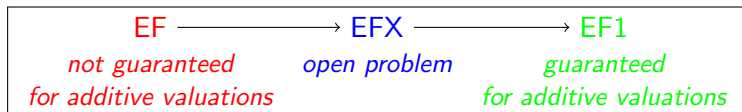
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>Ann</i>	15	3	2	2	6
<i>Bob</i>	7	5	5	5	7
<i>Carol</i>	20	3	3	3	3

- ▶ Between EF1 and EF: *envy-freeness up to any good* (EFX)
- ▶ the red allocation is EFX
- ▶ does an EFX allocation always exist?

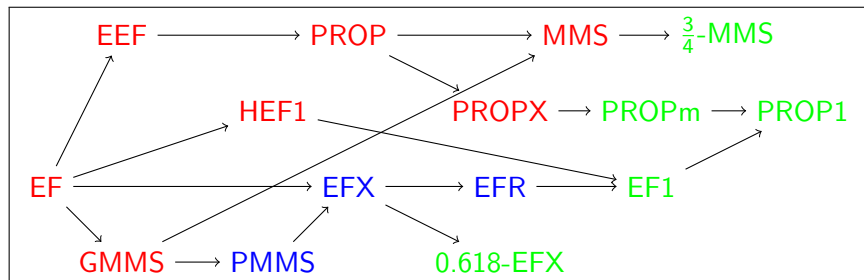
8. Fair Division of Indivisible Items

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
<i>Ann</i>	15	3	2	2	6
<i>Bob</i>	7	5	5	5	7
<i>Carol</i>	20	3	3	3	3

- ▶ Between EF1 and EF: *envy-freeness up to any good* (EFX)
- ▶ the red allocation is EFX
- ▶ does an EFX allocation always exist? Long-standing open problem



8. Fair Division of Indivisible Items



9. Iterated Voting

Plurality voting

4 voters $a \succ b \succ c \succ d \succ e$

3 voters $e \succ d \succ b \succ c \succ a$

2 voters $c \succ e \succ b \succ a \succ d$

2 voters $b \succ c \succ d \succ a \succ e$

9. Iterated Voting

Plurality voting

4 voters $a \succ b \succ c \succ d \succ e$

3 voters $e \succ d \succ b \succ c \succ a$

2 voters $c \succ e \succ b \succ a \succ d$

2 voters $b \succ c \succ d \succ a \succ e$

winner: a

9. Iterated Voting

Plurality voting

4 voters $a \succ b \succ c \succ d \succ e$

3 voters $e \succ d \succ b \succ c \succ a$

2 voters $c \succ e \succ b \succ a \succ d$

2 voters $b \succ c \succ d \succ a \succ e$

previous winner: a

winner: e

9. Iterated Voting

Plurality voting

4 voters $a \succ b \succ c \succ d \succ e$

3 voters $e \succ d \succ b \succ c \succ a$

2 voters $c \succ e \succ b \succ a \succ d$

2 voters $b \succ c \succ d \succ a \succ e$

previous winner: e

winner: b

Chances are that we have reached convergence.

9. Iterated Voting

4 voters	$a \succ b \succ c \succ d \succ e$	$a \succ b \succ c \succ d \succ e$
3 voters	$e \succ d \succ b \succ c \succ a$	$e \succ d \succ b \succ c \succ a$
2 voters	$c \succ e \succ b \succ a \succ d$	$c \succ e \succ b \succ a \succ d$
2 voters	$b \succ c \succ d \succ a \succ e$	$b \succ c \succ d \succ a \succ e$
winner	a	b

- ▶ voting rule + voter behaviour model $\rightarrow$ equilibrium reached?
- ▶ equilibria sometimes of better quality than sincere outcomes
- ▶ behaviour model validated (or not) by experiments
- ▶ also experiments with multiple referenda

10. Communication Protocols, Communication Complexity

- ▶ **communication protocol** for a voting rule: conditional sequence of communication actions with eventual identification of the true outcome of the rule.
- ▶ **communication complexity** of a voting rule: worst-case number of bits to be communicated by the voters in the best protocol for the rule.

Quiz: n voters, m candidates; communication complexity of constant rule:

10. Communication Protocols, Communication Complexity

- ▶ **communication protocol** for a voting rule: conditional sequence of communication actions with eventual identification of the true outcome of the rule.
- ▶ **communication complexity** of a voting rule: worst-case number of bits to be communicated by the voters in the best protocol for the rule.

Quiz: n voters, m candidates; communication complexity of

constant rule: 0 dictatorship:

10. Communication Protocols, Communication Complexity

- ▶ **communication protocol** for a voting rule: conditional sequence of communication actions with eventual identification of the true outcome of the rule.
- ▶ **communication complexity** of a voting rule: worst-case number of bits to be communicated by the voters in the best protocol for the rule.

Quiz: n voters, m candidates; communication complexity of

constant rule: 0 dictatorship: $\log m$ plurality:

10. Communication Protocols, Communication Complexity

- ▶ **communication protocol** for a voting rule: conditional sequence of communication actions with eventual identification of the true outcome of the rule.
- ▶ **communication complexity** of a voting rule: worst-case number of bits to be communicated by the voters in the best protocol for the rule.

Quiz: n voters, m candidates; communication complexity of

constant rule: 0 dictatorship: $\log m$ plurality: $n \log m$
Borda:

10. Communication Protocols, Communication Complexity

- ▶ **communication protocol** for a voting rule: conditional sequence of communication actions with eventual identification of the true outcome of the rule.
- ▶ **communication complexity** of a voting rule: worst-case number of bits to be communicated by the voters in the best protocol for the rule.

Quiz: n voters, m candidates; communication complexity of

constant rule: 0	dictatorship: $\log m$	plurality: $n \log m$
Borda: $n \log(m!)$	pl. runoff:	

10. Communication Protocols, Communication Complexity

- ▶ **communication protocol** for a voting rule: conditional sequence of communication actions with eventual identification of the true outcome of the rule.
- ▶ **communication complexity** of a voting rule: worst-case number of bits to be communicated by the voters in the best protocol for the rule.

Quiz: n voters, m candidates; communication complexity of

constant rule: 0	dictatorship: $\log m$	plurality: $n \log m$
Borda: $n \log(m!)$	pl. runoff: $n(\log m + 1)$	STV: ?

10. Communication Protocols, Communication Complexity

Single transferable vote (STV)

33	a	⋻	b	⋻	c	⋻	d	⋻	e
16	b	⋻	d	⋻	c	⋻	e	⋻	a
3	c	⋻	d	⋻	b	⋻	a	⋻	e
8	c	⋻	e	⋻	b	⋻	d	⋻	a
18	d	⋻	e	⋻	c	⋻	b	⋻	a
22	e	⋻	c	⋻	b	⋻	d	⋻	a

33	a	⋻	b	⋻	d	⋻	e
16	b	⋻	d	⋻	e	⋻	a
3	d	⋻	b	⋻	a	⋻	e
8	e	⋻	b	⋻	d	⋻	a
18	d	⋻	e	⋻	b	⋻	a
22	e	⋻	b	⋻	d	⋻	a

33	a	⋻	d	⋻	e
16	d	⋻	e	⋻	a
3	d	⋻	a	⋻	e
8	e	⋻	d	⋻	a
18	d	⋻	e	⋻	a
22	e	⋻	d	⋻	a

33	a	⋻	d
16	d	⋻	a
3	d	⋻	a
8	d	⋻	a
18	d	⋻	a
22	d	⋻	a

$d \succ a \succ e \succ b \succ c$

10. Communication Protocols, Communication Complexity

A protocol for STV:

step 1 voters send their most preferred candidate to the central authority (C)

↪ $n \log m$ bits

10. Communication Protocols, Communication Complexity

A protocol for STV:

step 1 voters send their most preferred candidate to the central authority (C)

↪ $n \log m$ bits

step 2 let x be the candidate to be eliminated. All voters who had x ranked first receive a message from C asking them to send the name of their next preferred candidate. There were at most $\frac{n}{m}$ such voters

↪ $2 \frac{n}{m} \log m$ bits

10. Communication Protocols, Communication Complexity

A protocol for STV:

step 1 voters send their most preferred candidate to the central authority (C)

↪ $n \log m$ bits

step 2 let x be the candidate to be eliminated. All voters who had x ranked first receive a message from C asking them to send the name of their next preferred candidate. There were at most $\frac{n}{m}$ such voters

↪ $2 \frac{n}{m} \log m$ bits

step 3 similarly with the new candidate y to be eliminated. At most $\frac{n}{m-1}$ voters voted for y

↪ $2 \frac{n}{m-1} \log m$ bits

10. Communication Protocols, Communication Complexity

A protocol for STV:

step 1 voters send their most preferred candidate to the central authority (C)

↪ $n \log m$ bits

step 2 let x be the candidate to be eliminated. All voters who had x ranked first receive a message from C asking them to send the name of their next preferred candidate. There were at most $\frac{n}{m}$ such voters

↪ $2 \frac{n}{m} \log m$ bits

step 3 similarly with the new candidate y to be eliminated. At most $\frac{n}{m-1}$ voters voted for y

↪ $2 \frac{n}{m-1} \log m$ bits

etc.

10. Communication Protocols, Communication Complexity

A protocol for STV:

step 1 voters send their most preferred candidate to the central authority (C)

↪ $n \log m$ bits

step 2 let x be the candidate to be eliminated. All voters who had x ranked first receive a message from C asking them to send the name of their next preferred candidate. There were at most $\frac{n}{m}$ such voters

↪ $2 \frac{n}{m} \log m$ bits

step 3 similarly with the new candidate y to be eliminated. At most $\frac{n}{m-1}$ voters voted for y

↪ $2 \frac{n}{m-1} \log m$ bits

etc.

► $\text{total} \leq 2n \log m \left(1 + \frac{1}{m} + \frac{1}{m-1} + \dots + \frac{1}{2}\right) = \mathcal{O}(n(\log m)^2).$

10. Communication Protocols, Communication Complexity

A protocol for STV:

step 1 voters send their most preferred candidate to the central authority (C)

↪ $n \log m$ bits

step 2 let x be the candidate to be eliminated. All voters who had x ranked first receive a message from C asking them to send the name of their next preferred candidate. There were at most $\frac{n}{m}$ such voters

↪ $2 \frac{n}{m} \log m$ bits

step 3 similarly with the new candidate y to be eliminated. At most $\frac{n}{m-1}$ voters voted for y

↪ $2 \frac{n}{m-1} \log m$ bits

etc.

- ▶ total $\leq 2n \log m (1 + \frac{1}{m} + \frac{1}{m-1} + \dots + \frac{1}{2}) = \mathcal{O}(n(\log m)^2)$.
- ▶ Is this protocol optimal?
- ▶ best known lower bound 2005-2026: $\Omega(n \log m)$

10. Communication Protocols, Communication Complexity

A protocol for STV:

step 1 voters send their most preferred candidate to the central authority (C)

↪ $n \log m$ bits

step 2 let x be the candidate to be eliminated. All voters who had x ranked first receive a message from C asking them to send the name of their next preferred candidate. There were at most $\frac{n}{m}$ such voters

↪ $2 \frac{n}{m} \log m$ bits

step 3 similarly with the new candidate y to be eliminated. At most $\frac{n}{m-1}$ voters voted for y

↪ $2 \frac{n}{m-1} \log m$ bits

etc.

- ▶ total $\leq 2n \log m (1 + \frac{1}{m} + \frac{1}{m-1} + \dots + \frac{1}{2}) = \mathcal{O}(n(\log m)^2)$.
- ▶ Is this protocol optimal?
- ▶ best known lower bound 2005-2026: $\Omega(n \log m)$
- ▶ Communication complexity of STV: $\Theta(n(\log m)^2)$

10. Communication Protocols, Communication Complexity

- ▶ **communication protocol** for a voting rule: conditional sequence of communication actions with eventual identification of the true outcome of the rule.
- ▶ **communication complexity** of a voting rule: worst-case number of bits to be communicated by the voters in the best protocol for the rule.

Quiz: n voters, m candidates; communication complexity of

constant rule: 0	dictatorship: $\log m$	plurality: $n \log m$
Borda: $n \log(m!)$	pl. runoff: $n(\log m + 1)$	STV: $\mathcal{O}(n(\log m)^2)$

11. Collective Decision Making Datasets

Building & maintaining

Dataset for voting data:

PREFLIB.ORG

Other datasets: matching,
participatory budgeting

all open access

Exploiting

Gap between theory
and real-world instances?

Assessing the validity of
preference models

Learning/ discovering structure

11. Collective Decision Making Datasets

Building & maintaining

Dataset for voting data:
PREFLIB.ORG

Other datasets: matching,
participatory budgeting

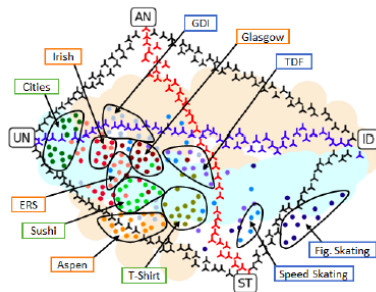
all open access

Exploiting

Gap between theory
and real-world instances?

Assessing the validity of
preference models

Learning/ discovering structure



“Map of real-world elections”

Source:

Boehmer, Brederick, Faliszewski,
Niedermeier & Szufa, 2021

Bibliography

- ▶ *Handbook of Computational Social Choice* (F. Brandt, V. Conitzer, U. Endriss, J. Lang, A. Procaccia, eds.). Cambridge University Press, 2016. Available online.
- ▶ *Trends in Computational Social Choice* (U. Endriss, ed.), 2017. Available online.

Contacting me: lang@lamsade.dauphine.fr